

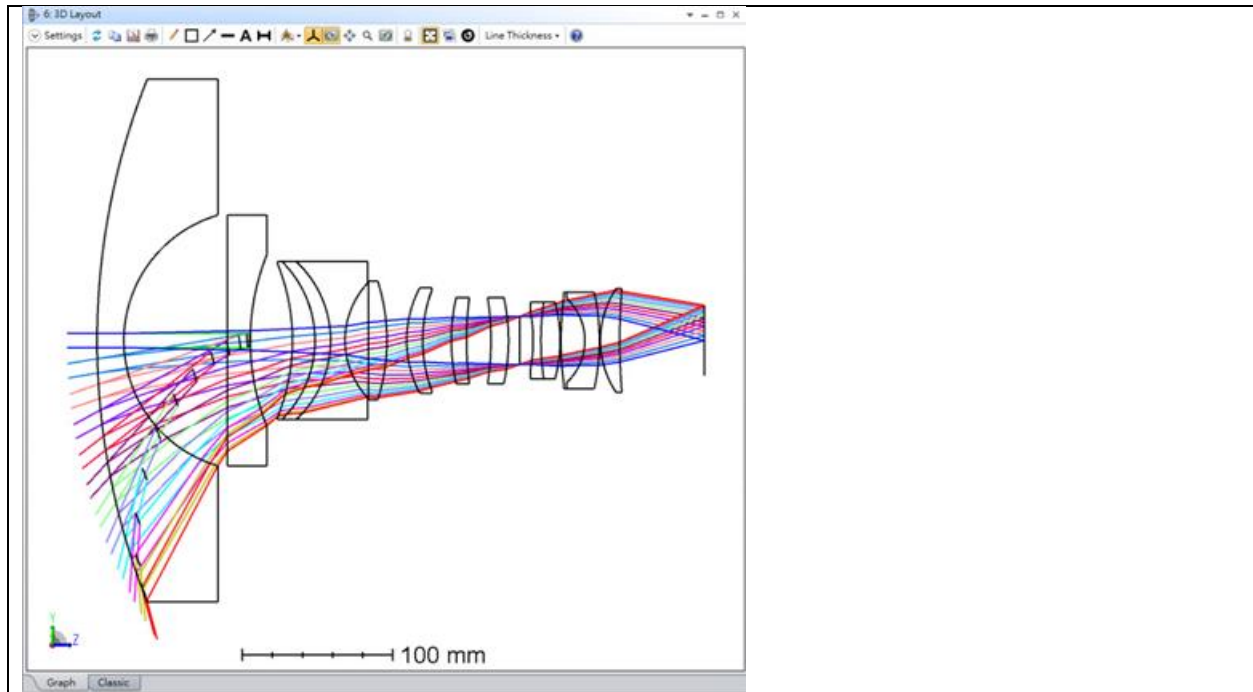
Title

ZPL_Plotting_Entrance_Pupil

Author

Michael Cheng - Zemax

Brief Description



Overview

For wide angle system, it can be interesting to visualize the position and the size of a real entrance pupil in different field angles. This macro will compute with real ray tracing real entrance pupil information for different field angles. It then adds the entrance pupil as a dummy surface and each “field” entrance pupil is defined into a configuration. The macro assumes axial-symmetric, only y field and the field angle in object is positive. Ray-Aiming should be turned on to correctly run this macro. It may only work with field type Angle and Object Height. How to use: Just run the macro, and open 3D Layout with showing central field and all configs.

Plugin Type

ZPL Macro

Language

ZPL

Updates

Date	Version	OpticStudio version	Comment
-	-	-	Creation

Instructions

1. Installation

To install the ZPL operand, copy the ZT_ENP.ZPL file under {Zemax}\Macros.

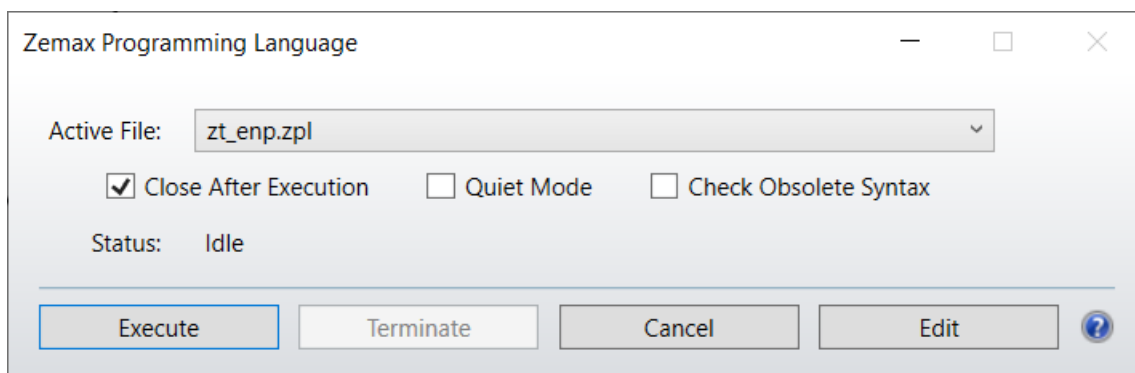
It is ready to be used.

2. How to use

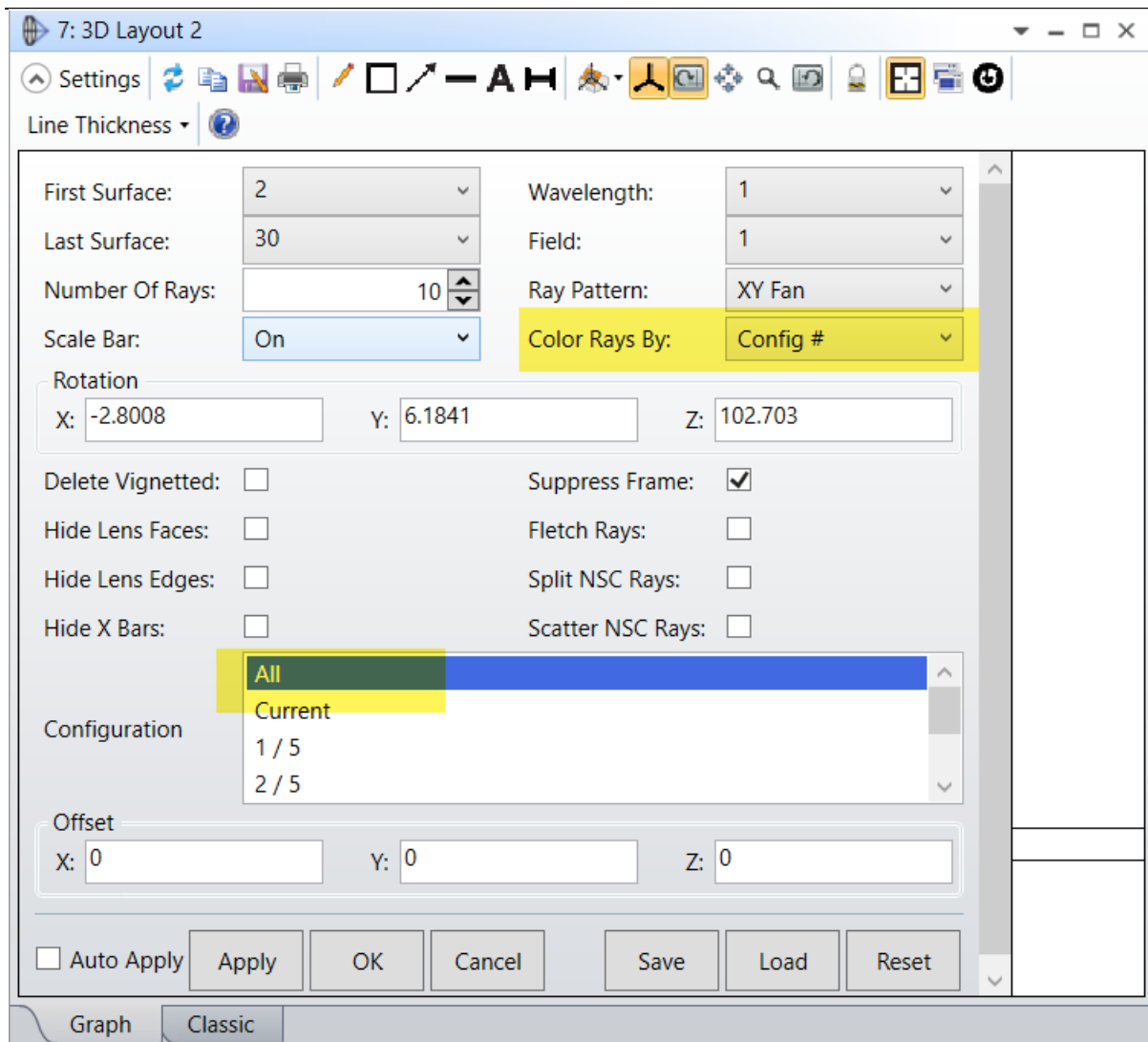
To use the macro, open a file that contains a doublet. For example

"{Zemax}\Samples\Sequential\Objectives\Wide angle lens 210 degree field.zos"

Run the macro:



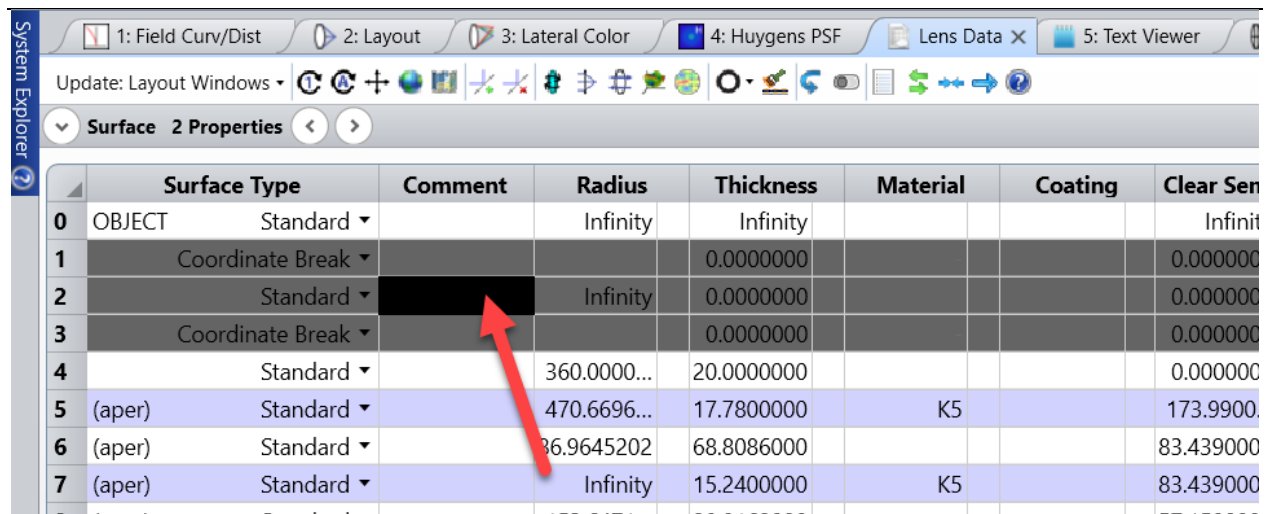
Open a new 3D layout and show all configurations:



Select surface 2:

Code Exchange

Zemax



The screenshot shows the Zemax System Explorer interface. The 'Surface 2 Properties' window is active, displaying a table of lens surfaces. A red arrow points to the 'Comment' column for Surface 2, which is currently blank. The table lists surfaces 0 through 7, including an object, coordinate breaks, and two aspherical surfaces (5 and 7) made of K5 material.

	Surface Type	Comment	Radius	Thickness	Material	Coating	Clear Sen
0	OBJECT Standard ▾		Infinity	Infinity			Infini
1	Coordinate Break ▾			0.0000000			0.000000
2	Standard ▾		Infinity	0.0000000			0.000000
3	Coordinate Break ▾			0.0000000			0.000000
4	Standard ▾		360.0000...	20.0000000			0.000000
5	(aper) Standard ▾		470.6696...	17.7800000	K5		173.9900.
6	(aper) Standard ▾		36.9645202	68.8086000			83.439000
7	(aper) Standard ▾		Infinity	15.2400000	K5		83.439000

It will show the real entrance pupil per field on the 3D layout:

Code Exchange

Zemax

